

Recognition of Hand Gripping Patterns Using Multi-Channel PPG Signals for Prosthetic Applications

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Abstract—Upper-limb prosthetic devices rely on accurate detection of user movement intentions to enable intuitive and functional control. While electromyography is widely used for this purpose, it presents limitations related to signal stability, susceptibility to interference, and system complexity. In this context, photoplethysmography (PPG) has emerged as a promising non-invasive alternative for sensing muscle activity, offering simpler instrumentation and improved robustness. This article aims to evaluate the feasibility of using an array of five PPG sensors to recognize three distinct hand gripping patterns for prosthetic hand control. PPG data were collected from five healthy subjects performing three hand grips: open, close, and precision grip. Four machine learning algorithms were evaluated for grip classification, and performance was compared in terms of classification accuracy. The k-nearest neighbors algorithm achieved the best performance, reaching an average accuracy of 99%. These results demonstrate the potential of multi-sensor PPG systems for reliable hand grip recognition, supporting their applicability in future prosthetic control strategies.

Index Terms—photoplethysmography, hand grip recognition, upper-limb prosthesis, machine learning.

I. INTRODUCTION

Prosthetic devices play a crucial role in restoring autonomy and functional independence for individuals with upper limb disabilities or limb loss. Their effectiveness largely depends on detecting the user's intended movements and translating them into control commands. In 2023, Brazil reported 645 hand and wrist amputation or disarticulation procedures [1].

Reliable recognition of hand movement patterns is therefore essential for the proper operation of underactuated prostheses, allowing users to perform daily tasks in a more natural and confident manner [2]. Hand movements are involved in approximately 85% of activities of daily living (ADLs) [3], which include different grasp types such as power grip (35%), precision grip (30%), and lateral grip (20%) [4].

The analysis of muscle activity is a key aspect in several application domains, including human-computer interaction, sports science, rehabilitation, and prosthetic control [5]. Muscle activity can be described as the process of contraction and relaxation of muscle fibers, driven by electrical impulses

generated by the central nervous system. These impulses initiate biochemical and mechanical processes within the muscle tissue, resulting in either static conditions, in which muscle length remains constant, or dynamic conditions, in which muscle length changes during movement [6].

Electromyography (EMG) is one of the most widely used techniques for measuring muscle activity, as it captures the electrical signals generated during muscle contractions through surface or intramuscular electrodes [7]. Although EMG is well established in both clinical and research settings, it presents several limitations. The acquired signals are sensitive to electromagnetic interference, which can compromise measurement reliability [8]. In addition, achieving low skin-electrode impedance requires careful skin preparation, such as cleaning with alcohol, which may be inconvenient and time consuming [9]. The signal acquisition and processing stages also demand relatively complex electronic systems, increasing both system cost and implementation effort [10].

As an alternative to EMG-based approaches, photoplethysmography (PPG) has gained attention as a non-invasive optical sensing technique. PPG measures volumetric changes in blood within peripheral tissues by exploiting the light absorption characteristics of blood in biological tissues and vessels [11]. Compared to EMG, PPG presents several advantages, including reduced susceptibility to electromagnetic interference and the absence of issues related to skin-electrode contact. By selecting suitable wavelengths within the near-infrared range, optical signals can effectively penetrate superficial tissues [12]. Furthermore, PPG systems typically rely on simpler electronic components, which can facilitate integration into wearable and low-cost sensing platforms [13].

Despite these advantages, PPG also presents important limitations. The acquired signals can be affected by ambient light interference and are highly sensitive to motion artifacts, which may compromise signal quality in dynamic conditions. In addition, signal characteristics may vary depending on factors such as sensor placement and individual physiological differences, including skin properties. In this study, some of these effects were mitigated through controlled acquisition

conditions and sensor design; however, they remain important considerations for real-world applications.

Previous studies have demonstrated the feasibility of PPG for various biomedical applications, including the detection of muscle activity [14] and hand gesture recognition. In particular, the recognition of nine distinct hand gestures using PPG sensors has been reported with an average classification accuracy of 89% [15]. However, despite these promising results, no studies were identified that apply PPG signals to the real-time control of prosthetic hands. This stands in contrast to Near-Infrared Spectroscopy (NIRS), a closely related optical technique, which has already been investigated in several works for muscle monitoring and prosthetic control applications [16]–[19].

Considering the limited number of studies, to the best of the authors' knowledge, exploring the use of PPG signals for prosthetic hand control, this work aims to advance the application of optical sensing in this domain. Specifically, this article investigates the use of an array of five PPG sensors to recognize three distinct hand gripping patterns, targeting functionalities commonly required in ADLs. This study builds upon previous work reported in [20], in which a single PPG sensor was used to successfully discriminate between hand opening and closing movements. By expanding both the number of sensors and the complexity of the recognized movements, the present study seeks to evaluate the feasibility of PPG-based sensing for more detailed and practical prosthetic hand control scenarios.

II. METHODOLOGY

This section describes the methodology used in the experiment, detailing data collection protocols, filtering techniques, feature extraction processes, and model training and evaluation methods.

A. Materials and Methods

The experimental setup employed three main hardware components: an ESP32 microcontroller, a PCA9548A I2C multiplexer, and five MAX30102 optical sensors, see Fig. 1. The ESP32 was responsible for data acquisition and communication with the host computer, while the PCA9548A multiplexer was used to enable time-multiplexed access to multiple MAX30102 sensors sharing the same I2C bus. This approach was necessary because the sensors operate with fixed I2C addresses, and the multiplexer allowed sequential reading of each sensor without address conflicts. The system was powered via the USB connection to the host computer, which supplied power to the ESP32 and, consequently, to the connected sensors.

Although each MAX30102 sensor was configured with a nominal sampling rate of 400 Hz, the effective acquisition rate is limited by the sequential reading process. The ESP32 cycles through all sensors, acquiring data from one sensor at a time. Considering channel switching delays, sensor read operations, and loop timing, the effective sampling rate per sensor is estimated to be in the range of approximately 10–25

Hz. As a result, measurements from different sensors are not acquired simultaneously, but with small temporal offsets. To mitigate this, signals are processed using synchronized analysis windows, and feature extraction is based on statistical descriptors (e.g., rolling mean, minimum, and maximum), which reduces sensitivity to small temporal misalignments. Given the relatively slow dynamics of the analyzed hand gestures, these offsets are not expected to significantly affect the classification task.

The five MAX30102 sensors were mounted on a custom-designed bracelet fabricated using a 3D printer. The bracelet incorporated an adjustable elastic strap, allowing it to be securely fitted to forearms with different circumferences. This configuration ensured consistent sensor–skin contact across subjects, improving signal quality while maintaining comfort during data acquisition.

To mitigate environmental and physiological sources of variability, the bracelet was manufactured using non-transparent material, reducing ambient light interference and ensuring stable optical coupling between the sensors and the skin. Additionally, to minimize the influence of skin tone variability, the system leverages wavelengths in the near-infrared range, as discussed in [12]. These wavelengths enable deeper penetration into biological tissues and reduce signal variability associated with differences in skin pigmentation.

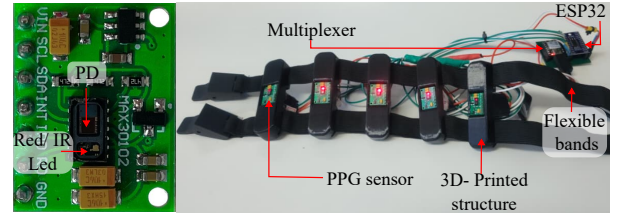


Fig. 1. Hardware components: a) PPG Sensor. b) Bracelet structure, multiplexer and microcontroller.

B. Data Collection Protocol

Data were collected from five healthy individuals, including three men and two women, with ages ranging from 20 to 28 years. To increase variability and better represent real usage conditions, the acquisition was performed over two different days, with multiple sessions per participant.

The sensors were positioned around the right forearm, with three sensors placed on the lower region and two on the upper region. The placement was adapted to each participant's anatomy, allowing small adjustments to ensure that all sensors maintained direct contact with the skin at an approximately perpendicular angle.

For each participant, data collection was divided into four sessions. In each day, two sessions were performed: one consisting of 5 seconds of baseline (hand open) followed by 5 seconds of hand closing, and another consisting of 5 seconds of baseline followed by 5 seconds performing a precision grip. Between sessions, short intervals were included to allow

repositioning of the bracelet and adjustment of the arm when necessary.

This protocol was designed to facilitate segmentation and post-processing while also introducing variability related to sensor repositioning. The objective was to classify hand states using data obtained from the red and infrared LEDs of the MAX30102 sensors. Fig. 2 shows the general setup and the experimental protocol.

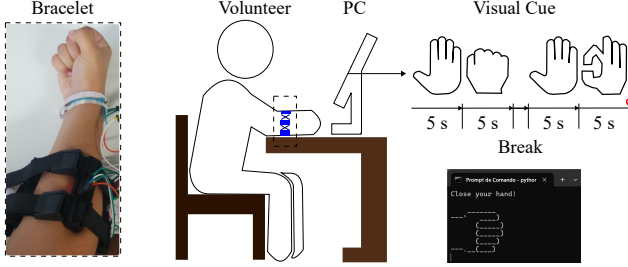


Fig. 2. Experimental protocol for gesture recognition and bracelet location.

C. Data Filtering

After acquisition, the recorded signals were preprocessed to improve consistency across sensors and participants. First, invalid samples (NaN values), typically caused by communication issues or incomplete readings, were removed from the dataset.

Next, a normalization step was applied using a Min–Max scaling function implemented in Python. This procedure scales each signal individually to a fixed range, reducing amplitude variations caused by sensor positioning, individual physiological differences, or hardware variability.

This preprocessing stage standardizes the signals and ensures that all sensors contribute comparably to the analysis, facilitating reliable feature extraction and improving the robustness of the subsequent machine learning models.

D. Feature Extraction

Feature extraction was performed using a sliding window approach applied to the red and infrared signals. The signals were sampled at an effective rate of approximately 20 Hz, and a window size of 30 samples was adopted, corresponding to a temporal duration of approximately 1.5 seconds.

Within each window, three statistical features were computed for each signal: rolling mean, rolling minimum, and rolling maximum. These features provide a compact representation of the signal behavior, capturing both its central tendency and variability over time.

This approach was chosen to maintain low computational complexity, making it suitable for real-time processing on resource-constrained hardware, while still preserving the relevant information required for accurate classification of hand movements.

E. Model Training

The dataset was composed of data from all participants, which were merged into a single dataset after the previous preprocessing steps.

The dataset was divided into training and testing subsets, with 80% used for training and 20% reserved for testing, using a fixed random state of 42 to ensure reproducibility. Based on recommendations from [21], lightweight machine learning models suitable for microcontroller deployment were selected, including Logistic Regression, K-Nearest Neighbors (k-NN), Decision Tree and Naive Bayes classifiers. These models were chosen for their low computational complexity and suitability for classification tasks in resource-constrained environments.

Hyperparameter optimization was performed using GridSearchCV from scikit-learn, which systematically explores predefined parameter grids to identify the best performing model configurations. A 5-fold cross-validation strategy was applied during this process, using only the training set.

F. Model Evaluation

Model performance was evaluated based on classification accuracy, which was used as the primary metric to assess the ability of each model to correctly recognize the hand grip classes. Accuracy was computed as the ratio of correctly classified samples to the total number of samples.

After hyperparameter tuning, the best model for each classifier was evaluated on the held-out test set (20%), which contains samples from multiple subjects, following a global modeling approach. Confusion matrices were also computed to provide a more detailed analysis of classification performance across classes.

To assess model consistency and reduce bias related to data partitioning, a 10-fold cross-validation strategy was adopted. In this approach, the dataset was divided into ten subsets, where each subset was used once as a validation set while the remaining subsets were used for training. The reported results correspond to the average accuracy and standard deviation across the folds.

III. RESULTS AND DISCUSSION

Fig. 3 illustrates the PPG signal acquired from the arrangement of sensors during the execution of the three evaluated hand gestures: open, close, and precision grip. Both the red and infrared channels are shown, each sensor captures distinct signal patterns associated with different hand grips, reflecting localized muscle activity. When combined, these complementary sensor-specific features enhance class separability, providing a more reliable basis for machine learning-based classification.

This qualitative analysis supports the use of multi-channel PPG sensing by showing how information from multiple sensors is jointly exploited to improve hand grip discrimination. Signals from all sensors are first normalized and then represented through statistical features, including rolling mean, rolling minimum, and rolling maximum. These features are

concatenated into a single feature vector and provided as input to the classifier.

The performance of the evaluated machine learning models was assessed using 10-fold cross-validation. Table I presents the average classification accuracy and standard deviation obtained for each classifier. Among the four evaluated algorithms, the k-NN classifier achieved the highest average accuracy, outperforming Decision Tree, Logistic Regression, and Gaussian Naive Bayes.

TABLE I
ACCURACY RESULTS PER MACHINE LEARNING MODEL.

Model	Accuracy (%)
Decision Tree	98.0 $\pm$ 0.01
Gaussian NB	57.0 $\pm$ 0.04
Logistic Regression	78.0 $\pm$ 0.03
k-NN	99.0 $\pm$ 0.01

The superior performance of k-NN indicates that the feature space extracted from the multi-channel PPG signals provides strong class separability for the considered hand gripping patterns. In contrast, Gaussian Naive Bayes exhibited the lowest accuracy and highest variability, suggesting that its assumption of feature independence is not well suited to the characteristics of PPG-based muscle activity signals.

Fig. 4 shows the classification accuracy across individual folds for each model. The k-NN classifier demonstrated consistently high performance across all folds, with accuracy values remaining close to 99 %, which explains the near-zero standard deviation reported in Table I. This stability suggests good generalization capability, despite the limited number of subjects included in the study.

Decision Tree also showed consistently stable performance across all folds, although with slightly lower accuracy compared to k-NN. Logistic Regression achieved moderate accuracy with limited variability, while Gaussian Naive Bayes presented both lower performance and greater fluctuation across folds, reinforcing its reduced suitability for this application.

To further analyze classification behavior, a confusion matrix was computed for the k-NN classifier, as it yielded the best overall performance. The confusion matrix, shown in Fig. 5, reveals high correct classification rates along the diagonal for all three hand grips (open, close, and precision) indicating reliable discrimination among the evaluated movements.

Minor misclassifications were primarily observed between the close and precision grips, which is expected given the similarity in muscle activation patterns required for these actions. Importantly, no systematic bias toward a specific class was observed, suggesting balanced recognition performance across grip types. This characteristic is particularly relevant for prosthetic control applications, where consistent behavior across different commands is essential for user confidence and usability.

It is important to note that, in the revised experimental protocol, data were collected across multiple sessions on different days to better represent real-world conditions, including sensor repositioning and environmental variability. Under

these conditions, the k-NN classifier maintained consistently high performance, whereas Logistic Regression exhibited a reduction in accuracy, reflecting its sensitivity to inter-session variability and distribution shifts. This behavior suggests that instance-based methods such as k-NN may be more robust for handling variability in PPG signals in practical applications.

Despite these promising results, an important limitation of the proposed approach is its high sensitivity to motion artifacts, which is inherent to PPG-based measurements. In the present study, data were collected under controlled and static conditions, which minimized motion-related noise and contributed to the high classification performance. However, in real-world scenarios involving dynamic movements, motion artifacts may significantly degrade signal quality and, consequently, classification accuracy. Addressing this limitation will be a key focus of future work, including the development of more robust signal processing techniques and evaluation under ecologically valid conditions.

IV. CONCLUSIONS AND FUTURE WORKS

The results indicate that multi-sensor PPG signals are suitable for recognizing hand grip patterns with high accuracy. In contrast to earlier studies based on a single PPG sensor and limited to hand opening and closing movements, the use of multiple sensors in this work enables the identification of more complex and functionally relevant grip types.

The performance of the k-NN classifier, together with the simplicity of the PPG hardware, demonstrates the feasibility of integrating this sensing approach into low-cost and wearable prosthetic systems. Compared to EMG-based solutions, the proposed approach avoids issues related to skin-electrode impedance and electromagnetic interference, which may contribute to improved robustness and long-term usability.

Despite these promising results, this study presents some limitations. Data were collected from a small group of healthy subjects, and further validation involving amputee users and real-time implementation is required. Additionally, an important limitation of the proposed approach is its high sensitivity to motion artifacts, which is inherent to PPG-based measurements. In the present study, data were collected under controlled and mostly static conditions, which minimized motion-related noise and contributed to the observed classification performance. However, in real-world scenarios involving dynamic movements, motion artifacts may significantly degrade signal quality and, consequently, classification accuracy.

Furthermore, for practical deployment in wearable systems, a detailed power analysis of the proposed setup is necessary, including the characterization of power consumption across system components and the evaluation of battery lifetime under continuous operation.

Future work will therefore focus on expanding the dataset, evaluating real-time performance, improving robustness to motion artifacts through advanced signal processing techniques, performing comprehensive power consumption analysis, and integrating the proposed approach into a complete prosthetic hand control framework.

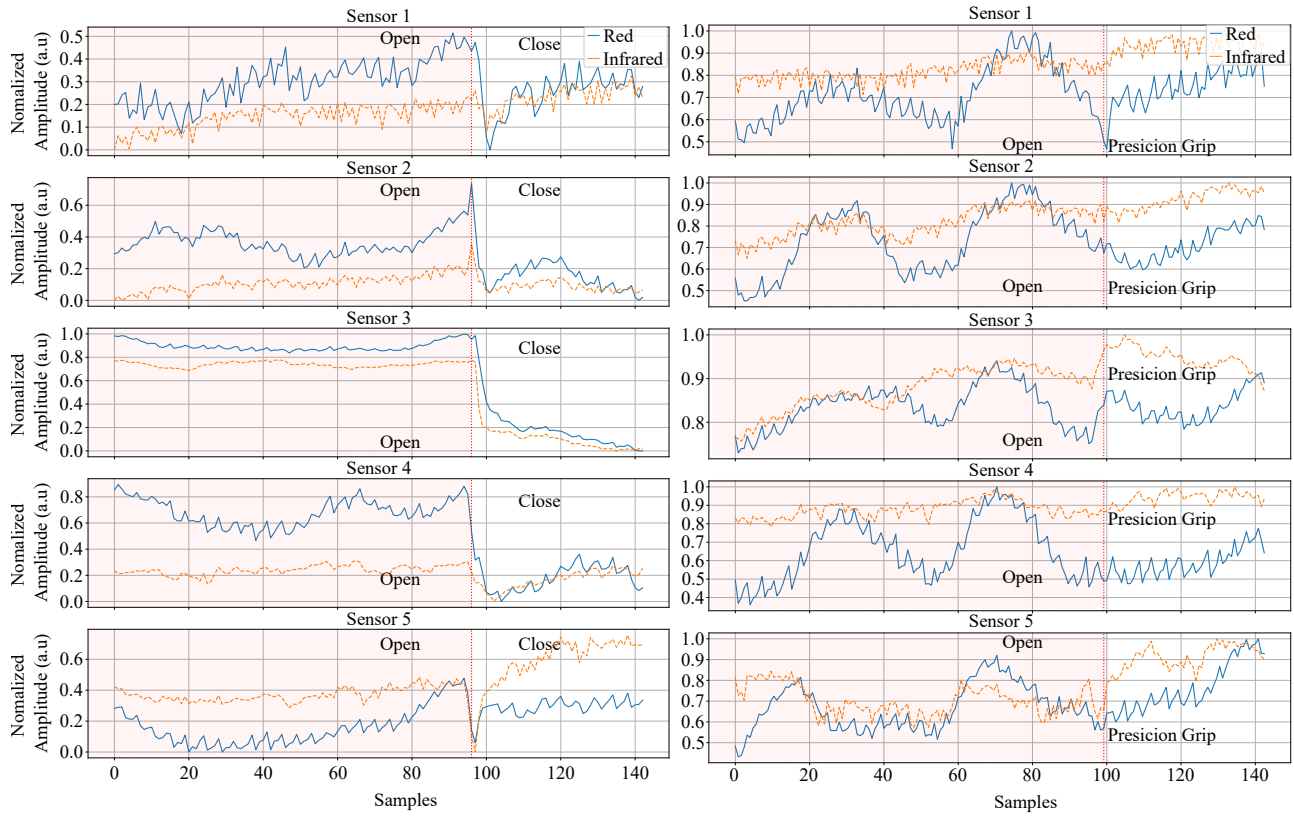


Fig. 3. Representative PPG signal acquired from the five sensors during the execution of open, close, and precision hand gestures. Red and infrared channels are shown, illustrating distinct signal patterns associated with each grip.

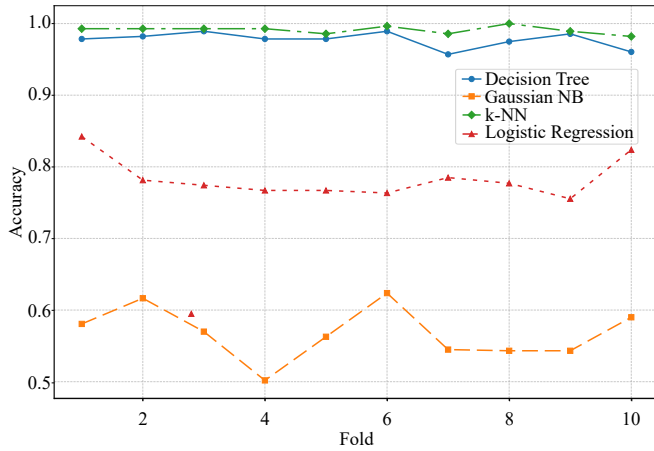


Fig. 4. Classification accuracy across individual folds for each model.

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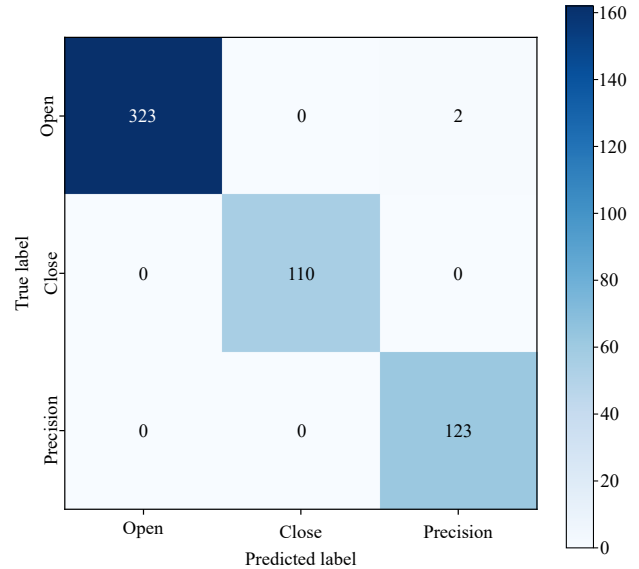


Fig. 5. Confusion matrix for k-NN algorithm.

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